LIMITS AT INFINITY

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Name: Solutions

Evaluate the following limits at infinity.

1. $\lim_{x\to\infty} \frac{3x-2}{2x+1}$

$$\lim_{x\to\infty} \frac{3x-2}{2x+1} = \lim_{x\to\infty} \frac{x(3-\frac{2}{x})}{x(2+\frac{1}{x})}$$

$$= \lim_{x\to\infty} \frac{3-2/x}{2+1/x}$$

$$= \frac{\lim_{x\to\infty} 3-2/x}{\lim_{x\to\infty} 2+1/x}$$

$$= \frac{3-0}{2+0}$$

$$= \boxed{\frac{3}{2}}$$

2. $\lim_{x\to\infty} \frac{4x^3+6x^2-2}{2x^3-4x+5}$

$$\lim_{x\to\infty} \frac{4x^3+6x^2-2}{2x^3-4x+5} = \lim_{x\to\infty} \frac{x^3(4+6/x-2/x^3)}{x^3(2-4/x^2+5/x^3)}$$

$$= \lim_{x\to\infty} \frac{4+6/x-2/x^3}{2-4/x^2+5/x^3}$$

$$= \frac{4+0-0}{2-0+0}$$

$$= \frac{4}{2}$$

$$= \boxed{2}$$

3. $\lim_{x\to\infty} \frac{\sqrt{2x^2+1}}{3x-5}$

$$\lim_{x\to\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to\infty} \frac{\sqrt{x^2(2+1/x^2)}}{x(3-5/x)}$$

$$= \lim_{x\to\infty} \frac{x}{x} \frac{\sqrt{2+1/x^2}}{3-5/x}$$

$$= \lim_{x\to\infty} \frac{\sqrt{2+1/x^2}}{3-5/x}$$

$$= \frac{\sqrt{2+0}}{3-0}$$

$$= \boxed{\frac{\sqrt{2}}{3}}$$

4. $\lim_{x\to\infty} \frac{x+3x^2}{4x-1}$

$$\lim_{x\to\infty} \frac{x^2(1/x+3)}{x(4-1/x)} = \lim_{x\to\infty} x \frac{(1/x+3)}{4-1/x}$$

$$= \boxed{\infty}$$

because

$$\lim_{x\to\infty} \frac{\frac{1}{x}+3}{4-\frac{1}{x}} = \frac{0+3}{4-0} = \frac{3}{4}.$$

5. $\lim_{x\to\infty} \frac{x^3-x}{x^2-6x+5}$

$$\lim_{x\to\infty} \frac{x^3-x}{x^2-6x+5} = \lim_{x\to\infty} \frac{x^3}{x^2} \frac{(1-1/x^2)}{(1-6/x+5/x^2)}$$

$$= \lim_{x\to\infty} x \frac{(1-1/x^2)}{(1-6/x+5/x^2)} = \infty$$

because

$$\lim_{x\to\infty} \frac{1-\frac{1}{x^2}}{1-6/x+5/x^2} = \frac{1-0}{1-0+0} = 1.$$

6. $\lim_{x\to\infty} \frac{x^4-3x^2+x}{x^3-x+2}$

$$\lim_{x\to\infty} \frac{x^4-3x^2+x}{x^3-x+2} = \lim_{x\to\infty} \frac{x^4}{x^3}\left(\frac{1-3/x^2+1/x^3}{1-1/x^2+2/x^3}\right)$$

$$= \lim_{x\to\infty} x \left(\frac{1-3/x^2+1/x^3}{1-1/x^2+2/x^3}\right)$$

$$= \infty$$

because $\lim_{x\to\infty} \frac{1-3/x^2+1/x^3}{1-1/x^2+2/x^3} = \frac{1-0+0}{1-0+0} = 1.$

7. $\lim_{x\to\infty} \frac{1-x^2}{x^3-x+1}$

$$\lim_{x\to\infty} \frac{1-x^2}{x^3-x+1} = \lim_{x\to\infty} \frac{x^2(1/x^2-1)}{x^3(1-1/x^2+1/x^3)}$$

$$= \lim_{x\to\infty} \frac{1}{x}\left(\frac{1/x^2-1}{1-1/x^2+1/x^3}\right)$$

$$= \left(\lim_{x\to\infty} \frac{1}{x}\right)\left(\lim_{x\to\infty} \frac{1/x^2-1}{1-1/x^2+1/x^3}\right)$$

$$= 0\left(\frac{0-1}{1-0+0}\right) = 0(-1) = \boxed{0.}$$

8. $\lim_{x\to\infty} \frac{1+x^4}{x^6+1}$

$$\lim_{x\to\infty} \frac{x^4}{x^6}\left(\frac{1/x^4+1}{1+\frac{1}{x^6}}\right) = \lim_{x\to\infty} \frac{1}{x^2}\left(\frac{\frac{1}{x^4}+1}{1+1/x^6}\right)$$

$$= 0\left(\frac{0+1}{1+0}\right) = \boxed{0.}$$

9. $\lim_{x\to\infty} \frac{x-2}{x^2+1}$

$$\lim_{x\to\infty} \frac{x-2}{x^2+1} = \lim_{x\to\infty} \frac{x}{x^2}\left(\frac{1-2/x}{1+1/x^2}\right)$$

$$= 0\left(\frac{1-0}{1+0}\right) = \boxed{0}$$

10. $\lim_{x\to\infty}\left(\sqrt{9x^2+x}-3x\right)$

$$\lim_{x\to\infty}\sqrt{9x^2+x}-3x = \lim_{x\to\infty}\frac{\sqrt{9x^2+x}-3x}{\sqrt{9x^2+x}+3x}\left(\sqrt{9x^2+x}+3x\right)$$

$$= \lim_{x\to\infty}\frac{9x^2+x-9x^2}{\sqrt{9x^2+x}+3x}$$

$$= \lim_{x\to\infty}\frac{x}{x\sqrt{9+1/x}+3x}$$

$$= \lim_{x\to\infty}\frac{1}{\sqrt{9+1/x}+3}$$

$$= \frac{1}{\sqrt{9+0}+3} = \frac{1}{3+3} = \boxed{\frac{1}{6}}$$

11. $\lim_{x\to\infty}\left(\sqrt{4x^2+3x}+2x\right)$

We don't need anything fancy here:

$$\lim_{x\to\infty}\sqrt{4x^2+3x}+2x = \boxed{\infty.}$$

Formally, if we choose any large number M, just choose $M<x$ so that

$$\sqrt{4x^2+3x}+2x > 2x > 2M > M.$$

12. $\lim_{x\to\infty}\left(x-\sqrt{x}\right)$

$$\lim_{x\to\infty} x-\sqrt{x} = \lim_{x\to\infty}\frac{x-\sqrt{x}}{x+\sqrt{x}}\left(x+\sqrt{x}\right)$$

$$= \lim_{x\to\infty}\frac{x^2-x}{x+\sqrt{x}}$$

$$= \lim_{x\to\infty}\frac{x^2(1-1/x)}{x(1+1/\sqrt{x})}$$

$$= \lim_{x\to\infty} x\left(\frac{1-1/x}{1+1/x^{1/2}}\right)$$

$$= \boxed{\infty}$$

because

$$\lim_{x\to\infty}\frac{1-1/x}{1+\frac{1}{x^{1/2}}} = \frac{1-0}{1+0} = 1.$$